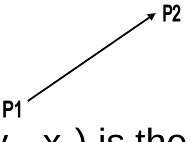
Computer Graphics

Mathematical Fundamentals

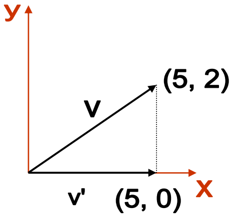
Vectors

- Another important mathematical concept used in graphics is the Vector



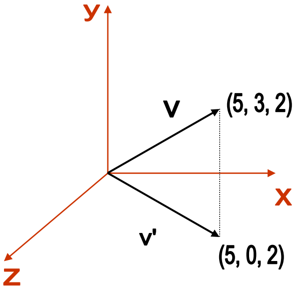
  - If $P_1 = (x_1, y_1, x_1)$ is the starting point and $P_2 = (x_2, y_2, z_2)$ is the ending point, then the vector $V = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$
  - This just defines *length and direction*, but *not position*

Vector Projections



- Projection of v onto the x-axis

Vector Projections



- Projection of v onto the xz plane

2D Magnitude and Direction

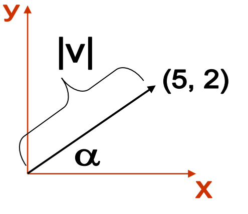
- The magnitude (length) of a vector:

  $|V| = \text{sqrt}(V_x^2 + V_y^2)$

  - Derived from the Pythagorean theorem
- The direction of a vector:

  $\alpha = \tan^{-1}(V_y / V_x)$

  - a.k.a. arctan or atan
  - α is angular displacement from the x-axis



3D Magnitude and Direction

- 3D magnitude is a simple extension of 2D

  $|V| = \text{sqrt}(V_x^2 + V_y^2 + V_z^2)$

- 3D direction is a bit harder than in 2D
  - Need 2 angles to fully describe direction
    - Latitude/longitude is a real-world example
  - Direction Cosines are often used:
    - α, β, and γ are the positive angles that the vector makes with each positive coordinate axes x, y, and z, respectively

$\cos\alpha = V_x / |V|$

$\cos\beta = V_y / |V|$ $\cos\gamma = V_z / |V|$

Vector Normalization

- "Normalizing" a vector means shrinking or stretching it so its magnitude is 1
  - a.k.a. creating unit vectors
  - Note that this does not change the direction
- Normalize by dividing by its magnitude

  $V = (1, 2, 3)$

  $|V| = \text{sqrt}(1^2 + 2^2 + 3^2) = \text{sqrt}(14) = 3.74$

  $V_{norm} = V / |V| = (1, 2, 3) / 3.74 =$
  $(1 / 3.74, 2 / 3.74, 3 / 3.74) = (.27, .53, .80)$

  $|V_{norm}| = \text{sqrt}(.27^2 + .53^2 + .80^2) = \text{sqrt}(.9) = .95$

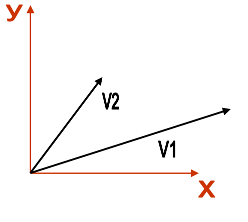
  - Note that the last calculation doesn't come out to exactly 1. This is because of the error introduced by using only 2 decimal places in the calculations above.

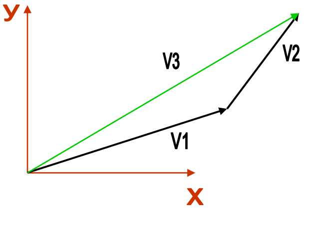
Vector Addition

- Equation:

  $V_3 = V_1 + V_2 = (V_{1x} + V_{2x} , V_{1y} + V_{2y} , V_{1z} + V_{2z})$

  $V_2 + V_1 = V_1 + V_2$ (commutative)
- Visually:



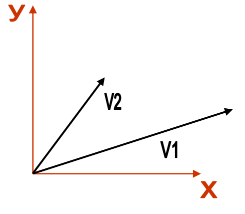


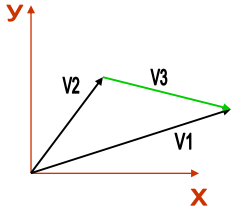
Vector Subtraction

- Equation:

  $V_3 = V_1 - V_2 = (V_{1x} - V_{2x}, V_{1y} - V_{2y}, V_{1z} - V_{2z})$

  Non-commutative -> Inverse Direction
- Visually:





Dot Product

(**Produto Interno**)

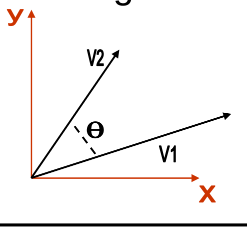
- The dot product of 2 vectors is a scalar

  $V_1 \cdot V_2 = (V_{1x} V_{2x}) + (V_{1y} V_{2y}) + (V_{1z} V_{2z})$
- Or, perhaps more importantly for graphics:

  $V_1 \cdot V_2 = |V_1| \; |V_2| \; \cos(\theta)$

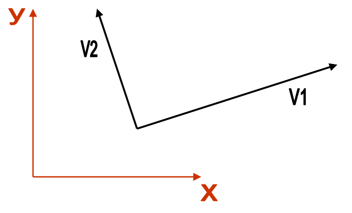
  where θ is the angle between the 2 vectors

  and θ is in the range $0 \leq \theta \leq \Pi$



Dot Product

- Why is dot product important for graphics?
  - It is zero if the 2 vectors are perpendicular
    - cos(90) = 0



Dot Product

- The Dot Product computation can be simplified when it is known that the vectors are unit vectors

  $V_1 \cdot V_2 = \cos(\theta)$

  because $|V_1|$ and $|V_2|$ are both 1

  - Saves 6 squares, 4 additions, and 2 sqrts

Cross Product (**Produto Vetorial**)

- The cross product of 2 vectors is a vector

$$V_1 \times V_2 = (\, V_{1y} V_{2z} - V_{1z} V_{2y} ,\; V_{1z} V_{2x} - V_{1x} V_{2z} ,\; V_{1x} V_{2y} - V_{1y} V_{2x} \,)$$

  – Note that if you are big into linear algebra there is also a way to do the cross product calculation using matrices and determinants

Cross Product

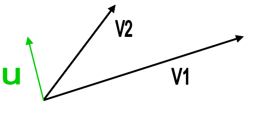
- Again, just as with the dot product, there is a more graphical definition:

$$V_1 \times V_2 = u \, |V_1| \; |V_2| \; \sin(\theta)$$

  where θ is the angle between the 2 vectors

  and θ is in the range $0 \leq \theta \leq \Pi$

  and u is the unit vector that is perpendicular to both vectors

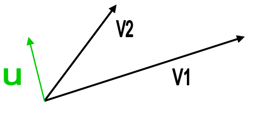


Cross Product

- Why u?
  - $|V_1|\ |V_2|\ \sin(\theta)$ produces a scalar and the result needs to be a vector

- The perpendicular definition leaves an ambiguity in terms of the direction of u
  - The direction of u is determined by the right hand rule

Cross Product

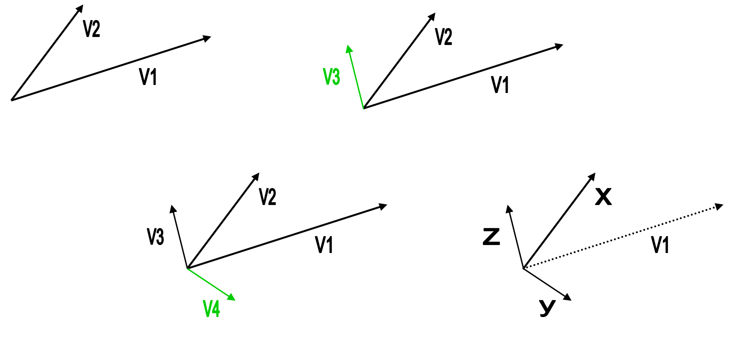


- Note that you can't take the cross product of 2 vectors that are parallel to each other
  - sin(0) = sin(180) = 0 → produces the vector (0, 0, 0)

Forming Coordinate Systems

- Cross products are great for forming coordinate system frames (3 vectors that are perpendicular to each other) from 2 random vectors
  - Cross V_1 and V_2 to form V_3
  - V_3 is now perpendicular to both V_1 and V_2
  - Cross V_2 and V_3 to form V_4
  - V_4 is now perpendicular to both V_2 and V_3
  - Then V_2, V_4, and V_3 form your new frame

Forming Coordinate Systems



- V_1 and V_2 are in the new xy plane